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Lab 1: Experiencing Blocking (Row-Level Lock)

```
156 -- 25/9/26
157 --Lab 1: Experiencing Blocking (Row-Level Lock)
158 BEGIN;
159
160 UPDATE film
161 SET rental_rate = 4.99
162 WHERE film_id = 1;
163
164 COMMIT;
165
166 SELECT film_id, rental_rate
167 FROM film
168 WHERE film_id = 1;
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 297 msec.

```
1 --25/9/26
2 --Lab 1:
3 BEGIN;
4
5 UPDATE film
6 SET rental_rate = 5.99
7 WHERE film_id = 1;
8
9 COMMIT;
10
```

Data Output Messages Notifications

Waiting for the query to complete...

156 -- 25/9/26
157 --Lab 1: Experiencing Blocking (Row-Level Lock)
158 BEGIN;
159
160 UPDATE film
161 SET rental_rate = 4.99
162 WHERE film_id = 1;
163
164 COMMIT;
165
166 SELECT film_id, rental_rate
167 FROM film
168 WHERE film_id = 1;

Data Output Messages Notifications
COMMIT
Query returned successfully in 96 msec.

4
5 UPDATE film
6 SET rental_rate = 5.99
7 WHERE film_id = 1;
8
9 COMMIT;
10

Data Output Messages Notifications
UPDATE 1
Query returned successfully in 41 secs 190 msec.

✓ Query returned successfully in 41 secs 190 msec. ✕

✓ Query returned successfully in 96 msec. ✕

1 --25/9/26
2 --Lab 1:
3 BEGIN;
4
5 UPDATE film
6 SET rental_rate = 5.99
7 WHERE film_id = 1;
8
9 COMMIT;
10

Data Output Messages Notifications
COMMIT
Query returned successfully in 60 msec.

✓ Query returned successfully in 60 msec. ✕

```
166 SELECT film_id, rental_rate
167 FROM film
168 WHERE film_id = 1;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1 Page No: 1

	film_id [PK] integer	rental_rate numeric (4,2)
1	1	5.99

Lab 2: Replicating Non-repeatable Read

```
194 -- Lab 2: Replicating Non-repeatable Read
195 -- Step 1
196 BEGIN;
197 SELECT rental_rate FROM film
198 WHERE film_id = 10;
199 -- Result: 4.99 (S-Lock released)
200
201 -- Step 3
202 SELECT rental_rate FROM film
203 WHERE film_id = 10;
204 -- NEWRESULT: 0.99!
205 COMMIT;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1 Page No: 1

	rental_rate numeric (4,2)
1	4.99

```

34  -- Lab 2: Replicating Non-repeatable Read
35  -- Step 2
36  BEGIN;
37  UPDATE film SET rental_rate = 0.99
38  WHERE film_id = 10;
39
40  COMMIT;
41  -- Modifies data and commits
42  -- while A is still active.

```

Data Output Messages Notifications

COMMIT

Query returned successfully in 54 msec.

```

201  -- Step 3
202  SELECT rental_rate FROM film
203  WHERE film_id = 10;
204  -- NEWRESULT: 0.99!
205  COMMIT;

```

Data Output Messages Notifications

COMMIT

Query returned successfully in 65 msec.

Lab 3: Creating a Deadlock

```

170  -- Lab 3: Creating a Deadlock
171  -- Step 1
172  BEGIN;
173  UPDATE film SET rental_rate = 2.99
174  WHERE film_id = 100;
175
176  -- Step 3
177  UPDATE film SET rental_rate = 3.99
178  WHERE film_id = 200;
179  -- Blocks (Waiting for B)

```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 65 msec.

```
12  -- Step 2
13  BEGIN;
14  UPDATE film SET rental_rate = 2.99
15  WHERE film_id = 200;
16
17  -- Step 4
18  UPDATE film SET rental_rate = 3.99
19  WHERE film_id = 100;
20  -- BOOM! ERROR: deadlock detected
```

Data Output	Messages	Notifications
UPDATE 1		
Query returned successfully in 57 msec.		

```
176  -- Step 3
177  UPDATE film SET rental_rate = 3.99
178  WHERE film_id = 200;
179  -- Blocks (Waiting for B)
```

Data Output	Messages	Notifications
Total rows: Waiting for the query to complete... 00:00:04.633 CRLF Ln 177, Col 1		

Data Output	Messages	Notifications
UPDATE 1 Query returned successfully in 30 secs 320 msec.	ERROR: deadlock detected Process 13888 waits for ShareLock on transaction 992; blocked by process 24600. Process 24600 waits for ShareLock on transaction 993; blocked by process 13888. SQL state: 40P01 Detail: Process 13888 waits for ShareLock on transaction 992; blocked by process 24600. Process 24600 waits for ShareLock on transaction 993; blocked by process 13888. Hint: See server log for query details. Context: while locking tuple (73,11) in relation "film"	✓ Query returned successfully in 30 secs 320 msec. ✕
Total rows: Query complete 00:00:30.320 CRLF Ln 177, Col 1	Total rows: Query complete 00:00:01.048 CRLF Ln 18, Col 1	

Lab 4: Preventing Race Conditions (FOR UPDATE)

```
181 -- Lab 4: Preventing Race Conditions (FOR UPDATE)
182 -- Step 1
183 BEGIN;
184 SELECT * FROM inventory
185 WHERE inventory_id = 10 FOR UPDATE;
186 -- Locks the row proactively
187
188 -- Step 3
189 INSERT INTO rental
190 (rental_date, inventory_id , customer_id, staff_id)
191 VALUES (NOW() , 10, 1, 1);
192 COMMIT;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

```
22 -- Lab 4: Preventing Race Conditions (FOR UPDATE)
23 -- Step 2
24 BEGIN;
25 SELECT * FROM inventory
26 WHERE inventory_id = 10 FOR UPDATE;
27 -- BLOCKS IMMEDIATELY
28
29 -- Step 4
30 -- Unblocks after A commits.
31 -- App logic detects it 's already rented!
32 ROLLBACK;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

```
188 -- Step 3
189 INSERT INTO rental
190 (rental_date, inventory_id , customer_id, staff_id)
191 VALUES (NOW() , 10, 1, 1);
192 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 70 msec.

```
29 -- Step 4
30 -- Unblocks after A commits.
31 -- App logic detects it 's already rented!
32 ROLLBACK;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 49 msec.